Multilabel Classification of Persuasion Techniques
Project Proposal for NLP Course, Winter 2026/27

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Abstract

Persuasion techniques play an important role in shaping public opinion. However, their automatic detection remains challenging, particularly for less-resourced languages such as Slavic languages. This project is conducted as part of the Slavic NLP 2027 Shared Task and focuses on sentence-level detection and classification of 25 persuasion techniques. Building on the paragraph-level annotations introduced in the Slavic NLP 2025 Shared Task, this project investigates methods for detecting and classifying persuasion techniques at the sentence level. The proposed work involves developing and evaluating NLP models, with a focus on multi-label classification. The main deliverable will be a reproducible classification pipeline, supported by experimental results and error analysis, providing insights into the effectiveness and limitations of different approaches to persuasion detection.

1 Introduction

Automatic detection of persuasion techniques can support the analysis of political and online discourse, including the study of propaganda and potentially misleading content. While benchmarks for English are well-established, fine-grained sentence-level detection for Slavic languages remains under-explored. To address this gap within the Slavic NLP 2027 Shared Task, this work evaluates multilingual pipelines across 25 persuasion techniques.

The remainder of this paper is structured as follows: Section 2 outlines the scientific goals and significance, Section 3 reviews related work in persuasion technique classification and multi-label modeling. Section 4 describes the datasets. Section 5 outlines our proposed experimental pipeline and Section 6 describes work plan.

2 Scientific Goal and Significance

The goal of this project is to build a multilingual NLP pipeline for the Slavic NLP 2027 Shared Task, addressing two core subtasks: (1) *binary classification* (detecting persuasion presence) and (2) *multi-label classification* (identifying among 25 fine-grained techniques).

The main research question is: To what extent can multilingual NLP models detect the presence of persuasion techniques and identify 25 fine-grained labels at the sentence level across Bulgarian, Polish and Russian, and how does performance vary across language–genre settings, label frequencies and unseen test languages?

3 Related Work and State of the Art

Automatic detection of persuasion and propaganda techniques has evolved from English-language news benchmarks towards multilingual, multi-genre and increasingly fine-grained settings, including sentence-level and multimodal classification. Below we review most significant works that trace this evolution, focusing on multi-label classification and Slavic languages. An overview of relevant benchmarks and systems, including their datasets, splits, methods and results, is provided in Appendix A (Table 4).

English benchmarks. NLP4IF-2019 (Da San Martino et al., 2019) introduced the Fine-Grained Propaganda Detection shared task, comprising two subtasks: sentence-level binary detection and span-level technique classification (18 techniques). The latter was won by a fine-tuned BERT token classifier with 0.25 F1 (Yoosuf and Yang, 2019), while sentence-level detection was won by (Mapes et al., 2019), achieving 0.63 F1. SemEval-2020 Task 11 (Da San Martino et al., 2020) reduced the taxonomy to 14 techniques;

a transformer ensemble with a BiLSTM-CRF (Morio et al., 2020) ranked first in span identification (0.52 F1) and third in classification (0.6173 micro-F1). Classification was won by (Jurkiewicz et al., 2020) with 0.6207 micro-F1.

Multi-label classification and imbalance. In SemEval-2021 Task 6 on English memes (Dimitrov et al., 2021), the top system reached 0.59 micro- but only 0.29 macro-F1, reflecting a long-tailed label distribution. On the same data, replacing binary cross-entropy with Asymmetric Loss (Ridnik et al., 2021) doubled macro-F1 in cross-validation, from 0.12 to 0.24 (Teimas and Saias, 2023). The 23-technique taxonomy of Piskorski et al. (2023b) was used in SemEval-2023 Task 3 (Piskorski et al., 2023a), covering nine languages including Polish and Russian. For Polish and Russian, the 2023 winner (Hromadka et al., 2023) was a single XLM-RoBERTa-large (XLM-R; Conneau et al. (2020)) trained on all languages, with thresholds calibrated separately for seen and surprise languages. It reached 0.43 and 0.39 micro-F1, but below 0.2 macro-F1. Bulgarian later reached only 0.11 span-level micro-F1 in CLEF-2024 (Piskorski et al., 2024) and 0.57 hierarchical F1 on memes (Dimitrov et al., 2024).

SlavicNLP 2025. The 2025 task (Piskorski et al., 2025) classified paragraphs from parliamentary debates (BG, HR, PL, SI) and Russian social media with 25 techniques. Detection was nearly saturated: the XLM-R-base baseline scored 0.83–0.94 F1 and was beaten only for HR, RU and SI, by at most 0.04, e.g., by contrastive debiasing (Ksiezniak et al., 2025). Classification stayed below 0.5 micro- and macro-F1 for all teams. Macro-F1 was led by two-pass chain-of-thought prompting of GPT-4o-mini and o4-mini (Jose and Greenstadt, 2025) (average ≈0.29), while fine-tuned XLM-R models mostly led micro-F1. The best fine-tuned system on macro-F1 (≈0.23) combined additional CLEF-2024 training data with per-language thresholds (Senichev et al., 2025), reaching 0.49 micro-F1 on Croatian. A hyperbolic hierarchical classifier over XLM-R-parla embeddings (Mochtak et al., 2024) (≈0.20) did not beat a flat XLM-R in its authors' own experiments (Brückner and Pecina, 2025), followed by LLM-generated explanations (Wang et al., 2025) (≈0.19).

Sentence level and positioning. The training corpus for the 2027 task (Piskorski et al., 2026) annotates sentences from Bulgarian and Polish debates and Russian Telegram posts. Its fine-tuned baselines reach 0.46/0.44/0.28 micro- but only 0.24/0.23/0.07 macro-F1. Guilt by Association and False Equivalence get zero F1 in all three languages, and even the "no technique" class reaches only 0.57–0.64 F1. Unlike at paragraph level, sentence-level detection is therefore not saturated, and classification is limited by rare techniques and cross-genre transfer (Russian was hardest in 2025). These baselines are the only published results for the 2027 setting we address.

4 Datasets and Data Statement

We currently plan to use two dataset sources for training and evaluating models for persuasion technique detection and classification: the primary Slavic NLP 2027 Shared Task dataset and an auxiliary English meme-text dataset derived from SemEval-2021 Task 6. Additional external datasets may be considered such as SemEval-2023 Task 3 and CLEF-2024 CheckThat! Task 3 data (Piskorski et al., 2023a, 2024), their status is RESTRICTED (access requested), and the project does not depend on them.

4.1 Primary Dataset: Slavic NLP 2027 Shared Task

- **Status:** `READY`
- **Origin & Acquisition:** Obtained from the official SlavicNLP GitHub repository. The corpus is described in (Piskorski et al., 2026). It contains Bulgarian and Polish parliamentary debate transcripts and Russian social media posts.
- **Size & Structure:** Table 1 presents the number of sentence-level instances divided into training, validation, and test splits.

Table 1: Sentence-level split sizes in the locally prepared SlavicNLP dataset.

Language	Train	Validation	Test	Total
Bulgarian	3,300	825	493	4,618
Polish	2,809	703	460	3,972
Russian	1,250	313	498	2,061
Total	**7,359**	**1,841**	**1,451**	**10,651**

- **Label Definition & Taxonomy:** The annotations follow a hierarchical taxonomy con-

sisting of 6 persuasion strategies and 25 fine-grained techniques. Each sentence can contain no persuasion technique or one or more technique labels. A list of all strategies and techniques, together with their descriptions, is provided in Appendix B.

- **Class Balance:** The dataset exhibits severe multi-label class imbalance. High-frequency categories such as *Loaded_Language* dominate the corpus, whereas techniques such as *Appeal_to_Time* represent sparse minority classes. The distribution of technique labels across all data splits is presented in Appendix C (Figure 2).

- **Licence:** The authors state the corpus is freely available for research purposes (Piskorski et al., 2026). Accompanying code: MIT.

4.2 Additional Dataset: SemEval-2021 Task 6 (English Memes)

- **Status:** `READY`

- **Origin & Acquisition:** Obtained from the official SemEval-2021 Task 6 repository (Dimitrov et al., 2021). We use the textual content of memes from the text-only multi-label classification subtask and preprocess the annotations to make them compatible with the primary dataset.

- **Size & Structure:** The dataset consists of English meme-text instances. Table 2 presents the number of instances in the locally prepared dataset, divided into training, validation, and test splits.

Table 2: Sentence-level split sizes in the locally prepared SemEval-2021 Task 6 dataset.

Language	Train	Validation	Test	Total
English	688	63	200	951

- **Label Definition & Preprocessing:** The original dataset contains 20 fine-grained persuasion technique labels. These labels are mapped to corresponding categories in the primary SlavicNLP taxonomy to make the annotations compatible for joint training. Descriptions of the techniques are provided in Appendix B.

- **Class Balance:** The preprocessed dataset exhibits a highly skewed label distribution. The detailed distribution of technique labels across all data splits is presented in Appendix C (Figure 3).

- **Licence:** The dataset repository states that the data are "free for general research use".

4.3 Data Statement

The dataset combines public news/social statements from the Slavic NLP 2027 Shared Task and English meme text from SemEval-2021 Task 6, both released for research use. They are public but contain personal data, such as names of speakers. We use them only for academic research and do not try to identify anyone. We do not redistribute raw data. Excerpts in the report have names replaced with placeholders. Annotations are subjective judgements of rhetorical techniques, not of truthfulness.

5 Solution Concept and Methodology

The design follows three findings from prior work (Section 3): fine-tuned multilingual encoders give the most stable results across languages, most errors concern rare techniques, and per-label threshold calibration and additional training data give the most consistent gains. We first build a *core system* trained only on the shared-task data, and then add auxiliary data in a fixed order, each step evaluated against the previous one (Figure 1).

5.1 Core System

Baselines. (B1) Character n-gram TF-IDF (term frequency–inverse document frequency) features with one-vs-rest logistic regression. (B2) XLM-R-base with a fixed threshold of 0.3, replicating the official 2025 baseline (Piskorski et al., 2025). (B3) The published sentence-level results of (Piskorski et al., 2026).

Model. An XLM-R encoder (Conneau et al., 2020) is fine-tuned jointly on all three languages, using the base model for development and XLM-R-large or XLM-R-parla (Mochtak et al., 2024), an XLM-R-large model further pre-trained on parliamentary proceedings, for the final system. The input is a single sentence, since test sentences may come without their surrounding text. The encoder feeds three heads: a binary head (ST1), a 25-label sigmoid head (ST2) and an 6-strategy head that exploits the taxonomy at no annotation cost.

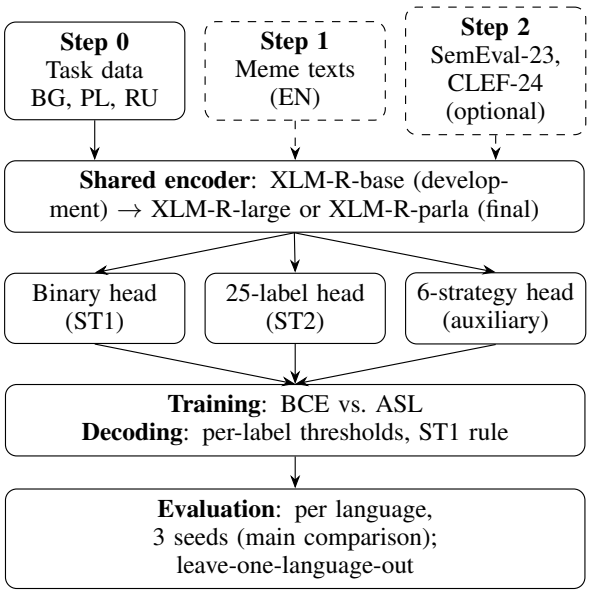


Figure 1: Proposed system. A multilingual encoder is fine-tuned on BG, PL and RU sentences of the shared-task corpus (Step 0), optionally extended with English meme texts (Step 1) and only if access is granted, sentence-projected span annotations from SemEval-2023 and CLEF-2024 (Step 2). The encoder feeds three heads: binary detection (Subtask 1, ST1), 25-technique classification (Subtask 2, ST2) and an 6-strategy head. We compare binary cross-entropy with Asymmetric Loss and tune per-label decision thresholds.

Compared techniques. (T1) *Loss*: BCE versus ASL (Ridnik et al., 2021), which improved macro-F1 on long-tailed persuasion data (Teimas and Saias, 2023). (T2) *Thresholds*: a fixed 0.3 versus per-label thresholds tuned on the validation split, pooled across languages (Hromadka et al., 2023; Senichev et al., 2025); labels with fewer than 5 validation positives fall back to a global threshold. (T3) *Auxiliary head*: with versus without the 6-strategy head. (T4) *ST1 decision rule*: binary head only, any predicted technique, or their disjunction, chosen on validation.

Unseen languages. In leave-one-language-out (LOLO) experiments we train on two languages and test on the third. Because the target languages also differ in genre (BG and PL debates, RU social media), held-out RU measures a combined language and genre shift. The LOLO folds also provide a second threshold set, tuned on held-out languages, which we apply to surprise languages in the official test set.

Auxiliary data. The auxiliary datasets (Section 4) are added to the best core configuration in a fixed order, and a step is kept only if it improves validation macro-F1. Step 1 adds the meme texts with mapped labels. Step 2, only if access is granted, adds SemEval-2023 and CLEF-2024 sentences labelled by span projection. Target techniques absent from a source taxonomy are masked in the loss rather than treated as negatives.

5.2 Evaluation

Metrics are fixed in advance. ST1: F1 of the positive class. ST2: macro-F1 over the 25 techniques (primary), micro-F1 and exact match ratio (secondary), and per-technique F1 grouped by training frequency (fewer than 20, 20–100, more than 100 sentences). For comparability with B3 we also report macro-F1 including the "no technique" class. Scores are reported per language on our test split. The main comparison (B2 versus the final system) is reported as mean$\pm$std over 3 seeds; other comparisons use a single seed because of the compute budget. On the official test set we report the score of the single submitted run.

Success deliverables. (D1) The final model outperforms B2 in macro-F1, averaged over 3 seeds. (D2) Full results of the LOLO experiments and of the T1–T4 comparisons are reported. (D3) A system submission with a reproducible pipeline.

6 Experimental and Work Plan

The project will be executed over the remainder of the semester, following the week-by-week plan outlined in Appendix D.

7 Mandatory Declarations

7.1 Contribution table

The individual contributions and workload distribution are detailed in Table 3

Table 3: Team Member Contributions

Initials	Hrs	Primary Contributions
A.K.	6	Exploratory Data Analysis (EDA), meme dataset preprocessing and introduction part of report.
M.M.	6	Part of the report related to Data (data description, analysis, data statements).
B.O.	6	Related works and Methodology part of the report.

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7.2 Declaration of LLM Use

Large Language Models (LLMs), specifically Anthropic Claude and OpenAI ChatGPT, were used to assist with academic text editing, generate data visualization utility functions, and support code development. All research design, data analysis, source code, and final content were independently reviewed and verified by the authors.

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A Extended Overview of Related Work

This appendix provides a detailed overview of previous persuasion detection benchmarks and published systems. The reported results are presented with their original evaluation metrics and should not be interpreted as a direct ranking across different datasets, languages, annotation units or task definitions.

Table 4: Overview of previous persuasion detection benchmarks and published systems.

Work	Dataset / Domain	Task / Annotation Unit	Evaluation Split	Method	Reported Results
(Da San Martino et al., 2019)	PTC, EN, news (18 techniques)	Binary detection (sentence-level, SLC)	Test, 86 articles (3,526 sentences)	Shared-task overview; baseline: logistic regression on sentence length	F1: baseline = 0.43; best system = 0.63
(Da San Martino et al., 2019)	PTC, EN, news (18 techniques)	Span identification + technique classification (span-level, FLC)	Test, 86 articles	Shared-task overview; baseline: random spans and random technique	Span-level F1: baseline = 0.00; best system = 0.25
(Yoosuf and Yang, 2019)	PTC, EN, news (18 techniques)	Technique classification (span-level)	Test, 86 articles	BERT token classifier	Span-level F1 = 0.25
(Mapes et al., 2019)	PTC, EN, news (18 techniques)	Binary detection (sentence-level)	Test, 86 articles	Attention transformer using BERT	Sentence-level F1 = 0.63
(Da San Martino et al., 2020)	SemEval-2020 Task 11, EN, news (14 techniques)	Binary detection (span-level, SI)	Test, 90 articles	Shared-task overview; baseline: random span generation	Span-level F1: baseline = 0.00; best system = 0.52
(Da San Martino et al., 2020)	SemEval-2020 Task 11, EN, news (14 techniques)	Technique classification (span-level, TC; gold spans given)	Test, 90 articles (1,790 spans)	Shared-task overview; baseline: logistic regression on span length	Micro-F1: baseline = 0.25; best system = 0.62
(Morio et al., 2020)	SemEval-2020 Task 11, EN, news (14 techniques)	Binary detection (span-level)	Test, 90 articles	Ensemble of PLM-based architectures with BiLSTMs, CRF, PoS/NE embeddings	Span-level F1 = 0.52
(Morio et al., 2020)	SemEval-2020 Task 11, EN, news (14 techniques)	Technique classification (span-level)	Test, 90 articles	Stacked ensemble of PLMs (BERT, GPT-2, XLNet, XLM, RoBERTa, XLM-R) with span/sentence FFN heads	Micro-F1 = 0.62
(Jurkiewicz et al., 2020)	SemEval-2020 Task 11, EN, news (14 techniques)	Binary detection (span-level)	Test, 90 articles	RoBERTa-large + CRF with three rounds of self-training on silver OpenWebText data	Span-level F1 = 0.49[a]
(Jurkiewicz et al., 2020)	SemEval-2020 Task 11, EN, news (14 techniques)	Technique classification (span-level)	Test, 90 articles	Context-aware ensemble with silver-data augmentation and reweighting	Micro-F1 = 0.6207[a]
(Dimitrov et al., 2021)	SemEval-2021 Task 6, EN, memes (20 text techniques)	Multi-label technique classification (meme text, ST1)	Test, 200 memes	Shared-task overview; baseline: majority class (Loaded Language)	Micro-F1: baseline = 0.37; best = 0.59
(Dimitrov et al., 2021)	SemEval-2021 Task 6, EN, memes (20 text techniques)	Span identification + technique classification (meme text, ST2)	Test, 200 memes	Shared-task overview; baseline: random spans and labels	Span-level F1: baseline = 0.01; best = 0.48
(Teimas and Saias, 2023)	SemEval-2021 Task 6, EN, memes (20 text techniques)	Multi-label technique classification (meme text, ST1)	Test, 200 memes	DistilBERT + CNN, Asymmetric Loss	Micro-F1 = 0.55

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Table 4 (continued)

Work	Dataset / Domain	Task / Annotation Unit	Evaluation Split	Method	Reported Results
(Dimitrov et al., 2024)	SemEval-2024 Task 4, memes; EN train/val/dev/test, BG/MK/AR test only (20 text techniques, hierarchical)	Hierarchical multi-label technique classification (meme text, ST 1)	Test: EN 1,500; BG 436; MK 259; AR 100 memes	Shared-task overview; baseline: majority class (Smears)	Hierarchical F1 (EN/BG/MK/AR): baseline = 0.37/0.28/0.31/0.36; best = 0.75/0.57/0.51/0.48
(Piskorski et al., 2023b)	Multilingual news, EN/FR/DE/IT/PL/RU (23 techniques)	Binary detection (paragraph-level, all languages)	All 6 languages	XLM-R-large token-level multi-label model, fine-grained training, binary evaluation	Micro-F1 = 0.83
(Piskorski et al., 2023b)	Multilingual news, EN/FR/DE/IT/PL/RU (23 techniques)	Multi-label technique classification (sentence-level)	All 6 languages	XLM-R-large multi-label classifier trained on all languages	Micro-F1 = 0.35
(Piskorski et al., 2023b)	Multilingual news, EN/FR/DE/IT/PL/RU (23 techniques)	Multi-label technique classification (paragraph-level, fine-grained)	All 6 languages	XLM-R-large token-level multi-label model	Micro-F1 = 0.38
(Piskorski et al., 2023a)	SemEval-2023 Task 3, Multilingual news, EN/FR/DE/IT/PL/RU /KA/EL/ES (23 techniques)	Multi-label technique classification (paragraph-level, ST3)	Test, 438 articles	Shared-task overview; baseline: linear SVM on word uni- and bigrams	Micro-F1: Baseline = 0.22, best system = 0.42
(Hromadka et al., 2023)	SemEval-2023 Task 3, Multilingual news, EN/FR/DE/IT/PL/RU /KA/EL/ES (23 techniques)	Multi-label technique classification (paragraph-level, ST3)	Test, 438 articles	Fine-tuned XLM-RoBERTa-large	Micro-F1 = 0.42
(Piskorski et al., 2024)	CLEF-2024 CheckThat! Task 3, Multilingual news; EN/FR/DE/IT/PL/RU /KA/EL/ES AR/BG/EN/PT/SL (23 techniques)	Span identification + multi-label technique classification	Test: AR 1,527 docs, BG 100, EN 98, PT 104, SL 100 docs	Shared-task overview; baseline: zero-shot XLM-RoBERTa-base token classification; best official systems: Mela (AR; mBERT with pre-training transfer), UniBO (BG, EN, PT, SL; BERT alignment data aug. + sentence/token classifiers)	Micro-F1 (AR/BG/EN/PT/SL): baseline = 0.021/0.009/0.009/0.002/0.003; best official = 0.301/0.114/0.092/0.107/0.123
(Piskorski et al., 2024)	CLEF-2024 CheckThat! Task 3, Multilingual news; EN/FR/DE/IT/PL/RU /KA/EL/ES AR/BG/EN/PT/SL (23 techniques)	Span identification + multi-label technique classification	Test: AR 1,527 docs, BG 100, EN 98, PT 104, SL 100 docs	Organizers' post-competition system (PersuasionMultiSpan*): XLM-RoBERTa token-level multi-label classifier trained on SemEval-2023 corpus, 50% overlap sliding-window chunking	Micro-F1 (AR/BG/EN/PT/SL) = 0.028/0.132/0.078/0.132/0.153
(Piskorski et al., 2025)	SlavicNLP 2025, BG/HR/PL/SI parliamentary debates, RU social media (25 techniques)	Binary detection (paragraph-level)	Test, 206 docs	Shared-task overview; baseline: fine-tuned XLM-R-base	F1 (BG/HR/PL/RU/SI) = 0.88/0.94/0.90/0.83/0.85
(Piskorski et al., 2025)	SlavicNLP 2025, BG/HR/PL/SI parliamentary debates, RU social media (25 techniques)	Multi-label technique classification (paragraph-level)	Test, 206 docs	Shared-task overview; baseline: fine-tuned XLM-R-base, threshold 0.3	Micro-F1 = 0.16/0.44/0.24/0.02/0.27

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Table 4 (continued)

Work	Dataset / Domain	Task / Annotation Unit	Evaluation Split	Method	Reported Results
(Jose and Greenstadt, 2025)	SlavicNLP 2025	Binary detection (paragraph-level)	Test, 206 docs	o4-mini prompting with definitions of all 25 techniques	F1 = 0.82/0.83/0.82/0.73/0.622
(Jose and Greenstadt, 2025)	SlavicNLP 2025	Multi-label technique classification (paragraph-level)	Test, 206 docs	GPT-4o-mini / o4-mini, CoT prompting with confidence cue, few-shot, two-pass (best run per language)	Micro-F1 = 0.41/0.44/0.42/0.29/0.3
(Wang et al., 2025)	SlavicNLP 2025	Binary detection (paragraph-level)	Test, 206 docs	XLM-R with GPT-4o cross-lingual augmentation and LLM explanations (concatenation or dual-encoder cross-attention; best configuration per language)	F1 = 0.81/0.89/0.88/0.87/0.65
(Wang et al., 2025)	SlavicNLP 2025	Multi-label technique classification (paragraph-level)	Test, 206 docs	XLM-R-base/large with GPT-4o augmentation and LLM explanations (best configuration per language)	Micro-F1 = 0.34/0.29/0.41/0.3/0.2
(Brückner and Pecina, 2025)	SlavicNLP 2025	Binary detection (paragraph-level; derived from multi-label predictions)	Test, 206 docs	XLM-R-parla embeddings with hyperbolic GCN and HIE loss (best variant per language)	F1 = 0.87/0.94/0.89/0.86/0.86
(Brückner and Pecina, 2025)	SlavicNLP 2025	Multi-label technique classification (paragraph-level)	Test, 206 docs	Hierarchical node classification in hyperbolic space over XLM-R-parla embeddings (best variant per language and metric)	Micro-F1 = 0.21/0.36/0.25/0.14/0.17
(Ksiezniak et al., 2025)	SlavicNLP 2025	Binary detection (paragraph-level)	Test, 206 docs	XLM-R-base multitask content debiasing (supervised contrastive loss on GPT-4o-neutralised pairs, $\lambda = 0.3$); HR: logistic regression on Jina walking embeddings	F1 = 0.88/0.96/0.9/0.84/0.85
(Senichev et al., 2025)	SlavicNLP 2025	Binary detection (paragraph-level)	Test, 206 docs	Slavic-BERT fine-tuned on official data only (no augmentation), tuned threshold	F1 = 0.84/0.85/0.88/0.86/0.81
(Senichev et al., 2025)	SlavicNLP 2025	Multi-label technique classification (paragraph-level)	Test, 206 docs	XLM-R-large with added CLEF-2024 RU/PL data (BG/HR/PL/SI); Slavic-BERT with EN/RU/PL data and GPT-4.1 translations (RU); per-language thresholds	Micro-F1 = 0.34/0.49/0.41/0.19/0.32
(Piskorski et al., 2026)	Slavic PT corpus, BG/PL/RU	Binary detection of non-persuasive sentences ("Without any PT")	Test, 1460 sentences	Per-language fine-tuned encoders: XLM-R-large (BG), polish-roberta-8k (PL), ruRoBERTa-large (RU)	F1 = 0.64/0.63/0.57
(Piskorski et al., 2026)	Slavic PT corpus, BG/PL/RU	Multi-label classification of 6 coarse strategies (sentence-level)	Full corpus	Zero-shot LLM prompting; best overall Gemma-3-27B	Micro-F1 = 0.40/0.36/0.28

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Table 4 (continued)

Work	Dataset / Domain	Task / Annotation Unit	Evaluation Split	Method	Reported Results
(Piskorski et al., 2026)	Slavic PT corpus, BG/PL/RU	Multi-label fine-grained technique classification (sentence-level)	Test, 1460 sentences	Per-language fine-tuned encoders: XLM-R-large (BG), polish-roberta-8k (PL), ruRoBERTa-large (RU)	Micro-F1 = 0.46/0.44/0.28
(Mochtak et al., 2024)	ParlaSent: Parliamentary proceedings (BS, HR, SR, CZ, EN, SK, SL)[b]	Sentence-level sentiment regression on a 6-level scale mapped to 0–5	BCS-test and EN-test (2,600 sentences each)	XLM-R-parla (XLM-R-large further pre-trained on 1.72B words of parliamentary text), fine-tuned on multilingual ParlaSent	R^2 (BCS/EN) = 0.615/0.672; MAE = 0.705/0.675

[a] Official test score reported in (Da San Martino et al., 2020); the system paper reports only development and cross-validation results.
[b] Not a persuasion benchmark; included as the source of the parliamentary-domain encoder XLM-R-parla used by (Brückner and Pecina, 2025).

B Persuasion Strategy and Technique Taxonomy

This appendix presents the hierarchical taxonomy of persuasion techniques used in the SlavicNLP dataset. The taxonomy consists of six high-level persuasion strategies, each comprising a set of fine-grained techniques, resulting in 25 technique labels in total. Table 5 lists all strategies and their corresponding techniques, together with descriptions taken directly from (Piskorski et al., 2026).

Table 5: Two-level persuasion taxonomy. The descriptions are based on the definitions provided in (Piskorski et al., 2026).

Persuasion Strategies	**Persuasion Techniques**	**Description**
Attack on Reputation	Name Calling or Labelling	A form of argument in which loaded labels are directed at an individual or a group, typically in an insulting or demeaning way.
	Guilt by Association	Attacking an opponent or an activity by associating it with another group, activity, or concept that has sharply negative connotations for the target audience.
	Casting Doubt	Casting doubt on the character or the personal attributes of someone or something in order to question their general credibility or quality, rather than using a proper argument relevant to the topic.
	Appeal to Hypocrisy	The reputation of the target is attacked by charging them with hypocrisy or inconsistency.
	Questioning the Reputation	This technique is used to attack the reputation of the target by making strong negative claims about it, focusing on undermining its character and moral stature rather than relying on an argument about the topic.
Justification	Flag-Waving (appeal to patriotism)	Justifying or promoting an idea by appealing to the pride of a group or highlighting the benefits for that specific group.
	Appeal to Authority	Attempting to add weight to an argument, an idea or information by simply stating that a particular entity considered to be an authority is the source of the information.
	Appeal to Popularity	This technique gives weight to an argument or idea by justifying it on the basis that allegedly "everyone" (or the vast majority) agrees with it, or "nobody" disagrees with it.
	Appeal to Fear, Prejudice	This technique aims at promoting or rejecting an idea through the repulsion or fear the audience feels toward this idea or toward its alternative.
	Appeal to Values	This technique gives weight to an idea by linking it to values seen by the target audience as positive.
Distraction	Strawman	This technique consists in creating an illusion of refuting the argument of the opponent's proposition, while the real subject of the argument was not addressed or refuted, but instead replaced with a false one.
	Whataboutism	Attempt to discredit an opponent's position by charging them with hypocrisy without directly disproving their argument.
	Red Herring	This technique consists in diverting the attention of the audience from the main topic being discussed, by introducing another topic.
	Appeal to Pity	Evokes feelings of pity, sympathy, compassion or guilt in audience to distract it from focusing on evidence, rational analysis and logical reasoning.
Simplification	Causal Oversimplification	Assuming a single cause or reason when there are actually multiple causes for an issue.

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Persuasion Strategies	Persuasion Techniques	Description
	False Dilemma or No Choice	Sometimes called the either-or fallacy, a false dilemma is a logical fallacy that presents only two options or sides when there are actually many.
	Consequential Oversimplification	An argument or an idea is rejected and instead of discussing whether it makes sense and/or is valid, the argument affirms, without proof, that accepting the proposition would imply accepting other propositions that are considered negative.
	False Equivalence	A technique that attempts to treat scenarios that are significantly different as if they had equal merit or significance.
Call	Slogans	A brief and striking phrase that may include labeling and stereotyping.
	Conversation Killer	This includes words or phrases that discourage critical thought and meaningful discussion about a given topic.
	Appeal to Time	The argument is centered around the idea that the time has come for a particular action.
Manipulative Wording	Loaded Language	Use of specific words and phrases with strong emotional implications (either positive or negative) to influence and to convince the audience that an argument is valid.
	Obfuscation, Intentional Vagueness, Confusion	This fallacy uses words that are deliberately unclear, so that the audience may have its own interpretations.
	Exaggeration or Minimisation	This technique consists of either representing something in an excessive manner—by making things larger, better, worse or by making something seem less important or smaller than it really is.
	Repetition	The speaker uses the same word, phrase, story, or imagery repeatedly in the hope that the repetition will persuade the audience.

C Class Distribution in the Datasets

This appendix presents the distribution of fine-grained persuasion techniques in the two datasets used in our experiments: the primary SlavicNLP dataset and the auxiliary SemEval-2021 Task 6 dataset. The figures show the number of examples annotated with each technique, aggregated across the training, validation, and test splits.

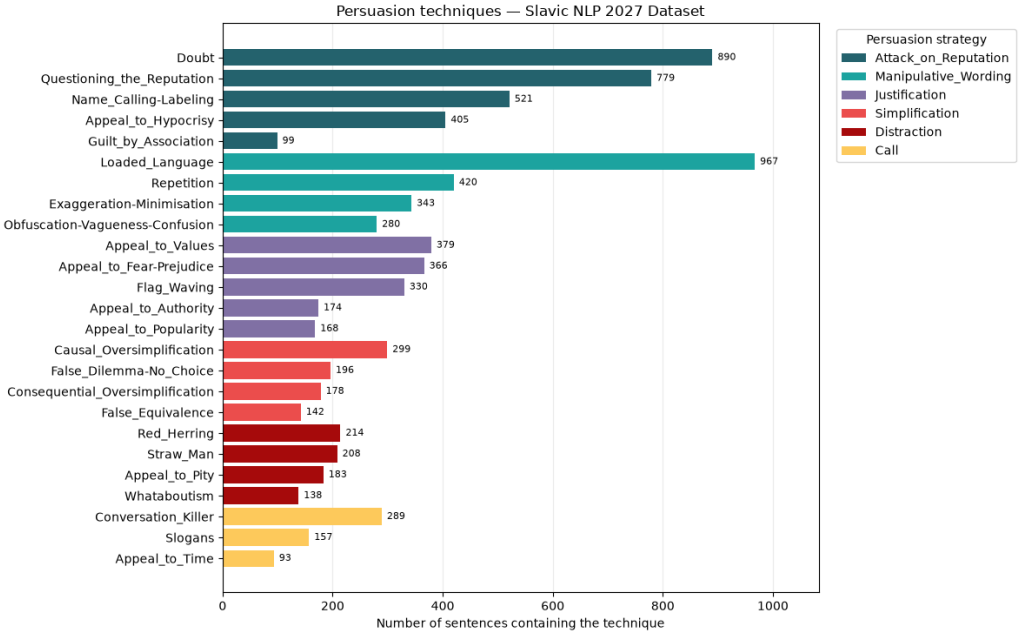


Figure 2: Distribution of the 25 fine-grained persuasion techniques in the SlavicNLP dataset. Counts are aggregated across the training, validation and test splits. The x-axis represents the number of examples, while the y-axis lists the corresponding persuasion techniques. Since both datasets use multi-label annotations, a single example may be assigned multiple techniques and therefore contribute to the counts of more than one label.

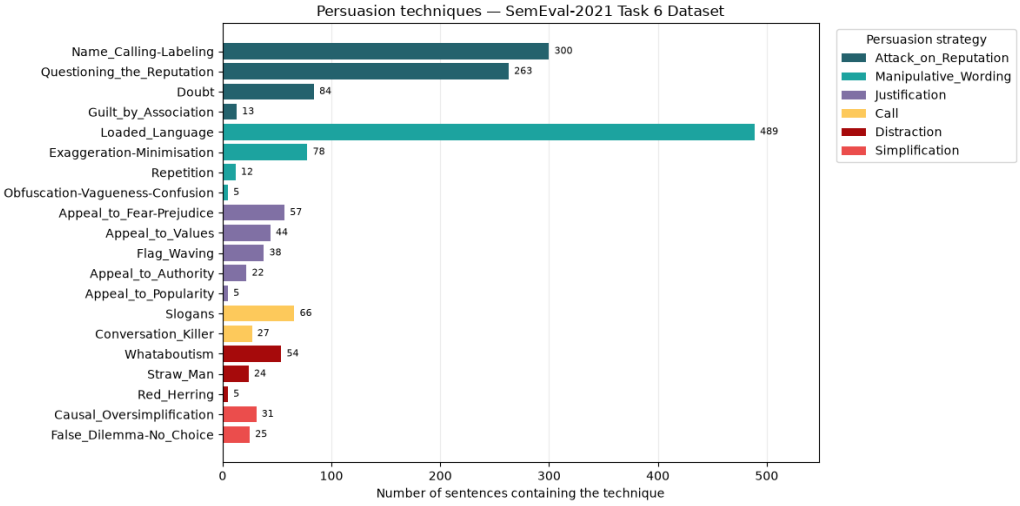


Figure 3: Distribution of persuasion techniques in the processed SemEval-2021 Task 6 dataset. Counts are aggregated across the training, validation and test splits. Labels are presented according to the mapping applied during preprocessing. The x-axis represents the number of examples, while the y-axis lists the corresponding persuasion techniques. Since both datasets use multi-label annotations, a single example may be assigned multiple techniques and therefore contribute to the counts of more than one label.

D Project Work Plan

The work plan outlines the main development stages, deliverables, task owners, and deadlines. It accounts for the course milestones as well as the official Slavic NLP 2027 Shared Task schedule.

Table 6: Project work plan, task allocation, and course and shared-task deadlines.

Week / Date	Task / Deliverable	Deadline / Milestone	Owner
5–6 (4–13 Nov 2026)	Baseline B1: character n-gram TF-IDF with one-vs-rest logistic regression		M.M.
	Baseline B2: XLM-R-base with a fixed decision threshold of 0.3		B.O.
	Baseline B3: compilation of published sentence-level results from (Piskorski et al., 2026)		A.K.
7 (18 Nov 2026)	Implementation of the multi-head architecture: shared encoder, binary detection head, 25-label classification head, and auxiliary 6-strategy head		M.M.
8–9 (25 Nov–2 Dec 2026)	Fine-tuning the XLM-R-base core model jointly on Bulgarian, Polish, and Russian sentence-level data		B.O.
	Step 1: mapping and integration of SemEval-2021 Task 6 English meme texts		A.K.
	Step 2: adaptation of SemEval-2023 and CLEF-2024 data through span projection, if access is granted		M.M.
	Comparison of binary cross-entropy (BCE) and Asymmetric Loss (ASL)	Course: 2 Dec 2026 (Phase 2)	B.O.
10–11 (9–18 Dec 2026)	Comparison of fixed and per-label decision thresholds, including the fallback to a global threshold for labels with fewer than 5 validation positives		M.M.
	Auxiliary 6-strategy head ablation and comparison of the three ST1 decision rules		A.K.
	Leave-one-language-out (LOLO) experiments: training on two languages and testing on the third		B.O.
	Final model experiments using XLM-R-large and/or XLM-R-parla		M.M.
	Shared Task registration	Shared Task: 18 Dec 2026	A.K.
12 (21 Dec 2026–7 Jan 2027)	Release of the official test data to registered participants	Shared Task: 21 Dec 2026	All
	Preparation of LOLO-based decision thresholds for application to surprise languages		A.K.
	Final evaluation: per-language ST1 and ST2 metrics, including macro-F1, micro-F1, exact match ratio, and per-technique F1 by training frequency		M.M.
	Three-seed comparison of B2 and the final system; single-seed evaluation of other experimental configurations		B.O.
	Inference on the official test data and submission of system responses	Shared Task: 7 Jan 2027	B.O.
13 (10 Jan 2027)	Review of the announced shared-task results	Shared Task: 10 Jan 2027	All
14 (13 Jan 2027)	Final course report and video; shared-task paper submission	Course: 13 Jan 2027 (Phase 3) Shared Task: 13 Jan 2027	All
Post-course (23 Jan 2027)	Shared-task paper notification	Shared Task: 23 Jan 2027	All
Post-course (30 Jan 2027)	Camera-ready shared-task paper submission	Shared Task: 30 Jan 2027	All

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